

Research Methodology

Systematic Literature Reviews (SLR)

Systematic Literature Reviews (SLR),

Survey Paper and Technical Paper

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Why do we need a Research Proposal

- **Why do we need a research proposal if the final goal is a technical paper?**
- A **research proposal** is like an **architectural blueprint** before building a house. You don't start coding or experimenting without a clear plan.

Why do we need a Research Proposal

1. Clarifies the Research Problem

- Students often say: *“We want to improve LLMs.”*
- That’s too broad! Improve **what**? Accuracy? Efficiency? Hallucination reduction?
- A proposal forces them to **narrow down the scope** and define a **specific research question**.

2. Defines the Methodology

- How will they improve the LLM?
 - Fine-tuning? Prompt engineering? Knowledge distillation?
- Proposal ensures they **plan the approach before implementation**.

3. Identifies Resources & Feasibility

- Do they have the **dataset**, **compute power**, and **time** to do it?
- Proposal helps avoid unrealistic projects.

4. Literature Gap Validation

- Proposal includes a **mini literature review** to show the idea is novel and not already solved.

5. Instructor Feedback

- You can **approve or refine** their idea early, preventing wasted effort.



Proposal Structure (Recommended)

- Title
- Introduction & Problem Statement
- Research Questions/Objectives
- Mini Literature Review (not full SLR)
- Proposed Methodology
- Expected Outcomes
- Timeline

Technical Paper vs Survey Paper

Feature	Technical Paper	Survey Paper
Goal	Present new research (method, model, experiment)	Summarize and analyze existing research
Content	Problem statement, methodology, experiments, results	Taxonomy, trends, comparisons, gaps
Contribution	Original solution or improvement	Comprehensive overview of a field
Example	“Improving LLM inference speed using LoRA”	“A Survey of Methods to Reduce Hallucination in LLMs”



Both are valuable and complementary

- Survey papers = Knowledge foundation
- Technical papers = Innovation on top of that foundation



Systematic Literature Review (SLR)



Systematic Literature Review

- **SLR** stands for **Systematic Literature Review**.
It's a **research method** used to collect, analyze, and synthesize all relevant studies on a specific research question in a **structured, transparent, and reproducible way**.
- **Key Features of SLR**
- **Purpose:**
To answer a well-defined research question by reviewing existing literature comprehensively.
- **Process:**
 - Define research question(s).
 - Develop a search strategy (databases, keywords).
 - Apply inclusion/exclusion criteria.
 - Assess quality of studies.
 - Extract and synthesize data.
- **Output:**
A paper that summarizes findings, identifies gaps, and suggests future research directions.
- SLR = Method
- Survey Paper = Output (which can be written using SLR method for rigor).



PRISMA Flow Diagram Steps

- **Identification**

- Records identified through database searching (e.g., Scopus, IEEE Xplore).
- Additional records identified through other sources (manual search, references).

- **Screening**

- Remove duplicates.
- Screen titles and abstracts for relevance.

- **Eligibility**

- Assess full-text articles for eligibility based on inclusion/exclusion criteria.

- **Included**

- Final studies included in the qualitative and/or quantitative synthesis.

- **Typical Numbers in Each Step**

- **Identification:** 2,000 papers found.
- **After duplicates removed:** 1,500.
- **After screening titles/abstracts:** 200.
- **After full-text eligibility check:** 50.
- **Final included:** 30.

Can write Survey Paper without using SLR?

- Yes, you **can** write a survey paper without using the full **SLR methodology**, but there are trade-offs:
 - **Survey Paper Without SLR**
 - **Approach:** Narrative or thematic review.
 - **Flexibility:** You choose which papers to include (based on relevance, citations, etc.).
 - **Pros:**
 - Faster to write.
 - Good for giving a broad overview of a field.
 - **Cons:**
 - May miss important studies.
 - Less reproducible and less rigorous.
- Increasingly less accepted in top-tier journals.
 - **Survey Paper With SLR**
 - Uses **systematic search and selection** (protocol-based).
 - **Pros:**
 - Comprehensive and unbiased.
 - Highly credible and publishable in top venues.
 - **Cons:**
 - Time-consuming.
 - Requires strict documentation.

Start to End in SLR

Before SLR

- You **already have a broad research area** (e.g., NLP, Computer Vision, AI in Healthcare).
- You may also have **some initial research questions** or themes.
- The goal is to **explore the literature systematically** to understand:
 - What has been done?
 - What gaps exist?
 - What methods and datasets are common?
- **Why Do SLR?**
- **To find research ideas:**
 - Identify **gaps** in existing work.
 - Spot **trends** and **future directions**.
- **To refine your research question:**
 - After SLR, you can make your question more specific and novel.
- **Key Point: SLR is not just a summary; it's a structured process** to map the research landscape and justify your future work.
- 💡 **Example:**
- Topic: *“Sentiment Analysis in Healthcare”*
- Initial Question: *“What methods are used for sentiment analysis in healthcare?”*
- After SLR: You might find a gap like *“Few studies use multimodal data for patient sentiment prediction”* → This becomes your research idea.

Three Different Things

- **LR** = A section in your research paper → quick summary of related work.
- **Survey** = A broad map of a research area → trends, taxonomy, challenges.
- **SLR** = A scientific method → strict process to collect & analyze all relevant studies.



Three Different Things

- **LR** = A section in your research paper → quick summary of related work.
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Three Different Things

Feature	Literature Review (LR)	Survey Paper	Systematic Literature Review (SLR)
Where it appears	Inside a research paper	Standalone paper	Standalone paper
Purpose	Give background & justify your research	Summarize a broad field	Answer a specific research question rigorously
Scope	Narrow (only related works)	Broad overview	Focused but comprehensive
Methodology	Narrative, subjective	Narrative, thematic	Structured, reproducible (protocol-based)
Example	“Related Work” section in your paper	“A Survey on Deep Learning for NLP”	“A Systematic Review of Transformer Models for Sentiment Analysis”



1. (SLR) Recommended Outline

- **Title & Abstract**
 - Clear, concise, and reflects the research question.
- **Introduction**
 - Background and motivation.
 - Research question(s) and objectives.
- **Methodology**
 - Search strategy (databases, keywords).
 - Inclusion/exclusion criteria.
 - Quality assessment process.
 - Data extraction and synthesis method.
- **Results**
 - PRISMA flow diagram (study selection process).
 - Summary tables of included studies.
 - Trends and patterns.
- **Discussion**
 - Key findings.
 - Research gaps.
 - Limitations of the review.
- **Conclusion & Future Work**
- **References**
- **Appendices**
 - Search strings, quality checklists.

2. (Survey Paper) Recommended Outline

- **Title & Abstract**
- **Introduction**
 - Overview of the domain.
 - Why this survey is needed.
- **Taxonomy / Classification**
 - Organize existing work into categories.
- **Detailed Review**
 - Discuss each category in depth.
 - Compare methods, datasets, metrics.
- **Trends and Challenges**
 - Emerging directions.
 - Open problems.
- **Conclusion**
 - Summary and future outlook.
- **References**
- *(Difference from SLR: Survey is broader, less rigid, and focuses on organizing and synthesizing knowledge rather than following a strict systematic protocol.)*

2. (Technical Paper) Recommended Outline

- **Title & Abstract**
- **Introduction**
 - Problem statement.
 - Motivation and contributions.
- **Related Work**
 - Brief review of existing approaches.
- **Proposed Methodology**
 - Model/algorithm details.
 - Architecture diagrams.
- **Experimental Setup**
 - Dataset description.
 - Evaluation metrics.
 - Implementation details.
- **Results and Analysis**
 - Quantitative results (tables, graphs).
 - Qualitative analysis (if applicable).
 - Comparison with baselines.
- **Discussion**
 - Strengths, limitations, and implications.
- **Conclusion & Future Work**
- **References**

Final Project

Why SLR in 1st Semester is Beneficial

- **Early Exposure to Research Process**

- Students learn how to search, filter, and critically analyze papers.
- Builds strong academic writing and referencing skills.

- **Helps in Thesis Topic Selection**

- By doing an SLR, students identify **gaps and trends** in their area of interest.
- This naturally leads to a **well-justified thesis problem** in later semesters.

- **Foundation for Proposal**

- The SLR can become the **literature review chapter** of their thesis.
- Saves time in the proposal and thesis writing stages.

- **Develops Research Rigor**

- Students learn systematic methods (PRISMA, inclusion/exclusion criteria).
- Prepares them for publishing in reputable venues.



Final Project Options

1. Technical Paper

- **Challenges:**

- Students need strong coding, ML/DL knowledge, and time for experiments.
- Risk: Many will struggle to produce meaningful results in 2 months.

- **When suitable:**

- For students with prior experience in ML/DL or strong programming background.

2. SLR (Systematic Literature Review)

- **Advantages:**

- Focuses on research skills: question formulation, systematic search, critical analysis.

- No heavy computation or deep coding required.
- Teaches academic writing and research rigor.

- **Feasible in 2 months:** Yes, if scope is well-defined and you provide a clear template.

- **Recommendation**

- **Make SLR the default final project** for all students.

- Allow **optional mini-technical experiments** for advanced students.

- This ensures:

- Everyone learns research methodology properly.
- No one is penalized for lack of deep learning expertise in the first semester.



Final Project Options - Group SLR

- **Why better for your case:**
 - Students can **divide tasks** (search, screening, synthesis, writing).
 - Encourages **collaboration and discussion**—important for research training.
 - Produces **higher-quality output** in limited time.
- **Ideal group size:** 3 students.
- **Assessment:**
 - Common report + individual contribution log (who did what).
 - Viva presentation for accountability.

